Anonymized Vector Discovery Through Semantic Atomization

Tom Tang
Edge Vector Foundation

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Abstract

We present a system for discovering privately stored data across a distributed network without exposing the underlying data, the data owner's identity, or the association between data fragments. Private data is fragmented into semantically meaningful segments, each independently converted into a vector embedding and published to a shared index tagged with a unique, one-time-use derived pseudonym generated from the owner's private key. No two fragments can be linked to each other or to the originating user. A querier searches by submitting a query vector and receives similarity-matched results identified only by unlinkable pseudonyms. The querier discovers the existence of similar data but cannot determine how fragments are associated, how many users contributed them, or access the raw data. An end-to-end encrypted relay enables the querier to contact the unknown data owner without revealing either party's identity.

1 Introduction

Existing systems force a binary choice: data is either private (and undiscoverable) or discoverable (and exposed). This paper presents a third option: data remains private and encrypted under the owner's control, while semantically meaningful fragments are published as vector embeddings under unlinkable pseudonyms, enabling similarity-based discovery without exposing content, identity, or fragment associations.

2 Background

2.1 The Discovery Problem

In existing systems, data exists in one of two states with respect to discoverability: public or private. Data that is stored privately (encrypted, access-controlled, or stored on local devices) is by definition undiscoverable by other parties. Data that is made discoverable must be exposed in a form that reveals its content, its owner, or both.

This binary creates a fundamental tension. A person who stores private photographs cannot benefit from others discovering semantically similar photographs without making those photographs public. A researcher with private medical data cannot discover that other researchers hold similar data without exposing the data itself. A creator with unpublished work cannot detect potential infringement of similar works without publishing.

2.2 Limitations of Prior Art

Existing approaches to privacy-preserving search and discovery suffer from several limitations:

Encrypted search systems (searchable encryption, homomorphic encryption) allow queries over encrypted data but typically require a shared encryption scheme between the data owner and querier, do not support semantic similarity search across unrelated parties, and impose substantial computational overhead.

Private information retrieval (PIR) protocols allow a user to query a database without revealing which record was accessed, but do not address the inverse problem: allowing others to discover that relevant data exists without revealing the data itself.

Federated learning enables model training across distributed data without centralizing it, but does not provide point-query discovery of specific similar data entries across the network.

Conventional pseudonymous systems assign a single pseudonym to each user, which enables linkage attacks: an adversary who observes multiple entries under the same pseudonym can correlate them, profile the user, and potentially deanonymize them through accumulated side information.

Centralized similarity search systems (reverse image search, plagiarism detection) require uploading data to a centralized index, surrendering control over the data and creating a surveillance-capable infrastructure.

Cryptocurrency stealth addresses (e.g., Monero) use per-transaction derived keys so that no two payments to the same recipient share an address, achieving transaction unlinkability. However, stealth addresses operate on financial transactions with exact-match lookups, not on semantic data with similarity search. They do not address the problem of fragmenting data for partial discoverability, do not involve vector embeddings or similarity matching, and do not provide a mechanism for anonymous communication between a querier and a data owner following discovery.

None of these approaches solve the core problem: enabling semantic discovery of private data across a distributed network of unrelated parties while preserving the privacy of the data content, the data owner's identity, and the association between data fragments.

3 System Overview

The system achieves privacy-preserving discovery through three mechanisms operating in combination:

1. **Content-aware fragmentation.** Private data is split into semantically meaningful fragments using content-type-specific splitting algorithms, such that each fragment carries independent semantic meaning but cannot be reassembled into the original data without access to the private store.

2. **Independent vectorization with per-fragment pseudonyms.** Each fragment is independently converted to a vector embedding and published to a shared network index. Each published fragment is tagged with a unique, one-time-use cryptographic pseudonym derived from the data owner's private key, such that no two fragments (whether from the same document or from different documents by the same user) share a pseudonym or can be linked by an external observer.

3. **Anonymous relay communication.** A querier who discovers a matching fragment can initiate communication with the unknown data owner through a relay system that forwards messages using the per-fragment pseudonym as an address, without revealing the identity of either party.

The combination of these three mechanisms enables a querier to discover that semantically similar private data exists somewhere in the network, determine a similarity score, but not determine which fragments belong to the same document, which documents belong to the same user, or access the underlying data. The querier can then privately contact the data owner to negotiate access, all without either party revealing their identity.

4 Architecture and Protocols

4.1 System Architecture Overview

The system comprises three layers (see Figure 1):

(a) A private storage layer in which each user's raw data resides under their exclusive control, protected by encryption and policy-enforcing access controls. Raw data never leaves this layer without explicit authorization.

(b) A public discovery layer consisting of a distributed vector index containing embedded fragments, each tagged with a unique derived pseudonym and associated metadata. This layer is publicly queryable. No access controls gate discovery queries.

(c) A relay communication layer that routes messages between queriers and data owners using per-fragment pseudonyms as addresses, without revealing the mapping between pseudonyms and real identities.

4.2 Content-Aware Fragmentation

When a user stores private data, the system fragments the data into semantically meaningful segments using content-type-specific algorithms. The fragmentation is designed so that each fragment carries independent semantic meaning sufficient for similarity matching, while the association between fragments (which fragments compose a single document) is known only to the data owner.

4.2.1 Text Fragmentation

For textual data (documents, essays, articles, emails, messages), the system performs sentence-level or paragraph-level splitting (see Figure 2). Each sentence or paragraph becomes an independent fragment. The splitting respects natural language boundaries so that each fragment is a coherent semantic unit.

For example, an essay of 50 sentences produces 50 independent fragments. Each fragment carries sufficient semantic content for similarity matching (a sentence about a specific medical condition will match queries about that condition), but no single fragment reveals the full content or structure of the essay.

4.2.2 Image Fragmentation

For image data (photographs, illustrations, scans), the system applies convolutional neural network (CNN) based segmentation to produce overlapping spatial segments (see Figure 3). Specifically:

(a) The image is processed through the initial convolutional layers of a CNN architecture to produce feature maps at multiple spatial scales.

(b) The CNN feature maps at multiple spatial scales are used to determine spatial boundaries in the original image. Corresponding image patches are then extracted from the original

image at these boundaries, with configurable overlap ratio (typically 10–25% of patch width) ensuring that features near boundaries appear in at least one adjacent fragment.

(c) Each extracted image patch is an independent fragment for subsequent vectorization.

(d) The overlap ensures that semantically significant features (faces, objects, text) that span region boundaries appear in at least one fragment in their entirety.

For example, a photograph containing a face, a building, and a landscape may produce 9–16 overlapping fragments. A fragment containing the face will match face-similarity queries; a fragment containing the building will match architectural queries. No single fragment reveals the full composition of the image, and the spatial relationship between fragments is known only to the data owner.

4.2.3 Audio Fragmentation

For audio data (recordings, music, speech), the system splits the audio into overlapping temporal segments at natural boundaries (silence detection, beat detection, or fixed-duration windows with overlap). Each segment is an independent fragment.

4.2.4 Structured Data Fragmentation

For structured data (medical records, financial records, tabular data), the system splits records into individual field-level or row-level fragments. Each fragment contains a single field or a small group of semantically related fields.

4.2.5 Code Fragmentation

For source code, the system uses syntax-aware parsing (e.g., tree-sitter) to split code into function-level or block-level fragments. Each fragment is a syntactically complete unit.

4.3 Independent Vectorization

Each fragment produced by the fragmentation step is independently converted to a vector embedding using a registered embedding transform:

$$T_{\text{embed}} : \text{Fragment} \to \mathbb{R}^d$$

The embedding transform is selected based on content type and is publicly identified by a content-addressed hash of its definition. All users who apply the same embedding transform produce vectors in the same embedding space, enabling cross-network similarity comparisons.

The embedding is designed to be practically irreversible: while exact reconstruction of the original fragment from the vector is computationally infeasible, the vector preserves semantic similarity relationships such that fragments with similar content produce vectors that are close in the embedding space.

Critical property: each fragment is embedded *independently*. The system does not produce a single embedding for the entire document, nor does it produce embeddings that encode relationships between fragments. The embedding of fragment i carries no information about fragment j, even when both originate from the same document.

4.4 Per-Fragment Derived Pseudonyms

Each embedded fragment is tagged with a unique cryptographic pseudonym before publication to the discovery layer (see Figure 4). The pseudonym generation works as follows:

4.4.1 Key Derivation

Let sk be the user's master private key. For each fragment i with content hash h_i, the system derives a unique public/private key pair:

$$(pk_i, sk_i) = \text{Derive}(sk, h_i)$$

where Derive is a hierarchical deterministic key derivation function (e.g., based on BIP-32 or HMAC-based derivation on an elliptic curve). Using the content hash h_i as the derivation path (rather than a sequential counter) eliminates the need for cross-device index coordination.

4.4.2 Security Properties

The derivation function satisfies the following properties:

(a) **Unlinkability.** Given two derived pseudonyms pk_i and pk_j, it is computationally infeasible for any party not possessing sk to determine whether pk_i and pk_j were derived from the same master key. The two pseudonyms are computationally indistinguishable from independently generated random keys to any probabilistic polynomial-time adversary not possessing the master private key sk.

(b) **Owner verifiability.** Given the master private key sk and any fragment content hash h_i, the owner can deterministically re-derive (pk_i, sk_i) and prove ownership of the fragment.

(c) **Non-transferability.** The derived private key sk_i can be used to sign messages (e.g., access grants, removal acknowledgments) that are verifiable against pk_i, without revealing sk or any other derived key.

(d) **Large-scale generation.** The derivation scheme supports generating at least 2^{256} unique pseudonyms from a single master key, which exceeds the number of atoms in the observable universe, and is limited only by the entropy of the content hashes.

If two different users independently fragment and publish identical content (e.g., the same sentence from a public document), they will derive different pseudonyms because the derivation function takes both the master private key sk and the content hash h_i as inputs: $pk_i = \text{Derive}(sk, h_i)$. Since each user has a unique master key, identical content produces distinct pseudonyms. However, if the same user publishes the same content twice, the deterministic derivation produces the same pseudonym, which is by design—it prevents duplicate entries and enables idempotent publication.

4.4.3 Publication

Each fragment is published to the public discovery layer as a tuple:

$$(pk_i,\ e_i,\ m_i)$$

where pk_i is the derived pseudonym, $e_i \in \mathbb{R}^d$ is the vector embedding, and m_i is public metadata (content type, embedding transform hash, timestamp). The raw fragment data is *not* published; it remains in the owner's private storage layer.

4.5 Similarity Discovery Protocol

A querier searches the network as follows (see Figure 5):

4.5.1 Query Submission

The querier generates a query vector $q \in \mathbb{R}^d$ by applying the same embedding transform to their query data (a photo, a sentence, a code snippet, etc.). The querier submits q and a similarity threshold ϵ to the public discovery index.

4.5.2 Index Search

The system searches the public vector index using approximate nearest neighbor algorithms (e.g., HNSW, IVF) and returns all matching entries:

$$\text{Result}(q, \epsilon) = \{(pk_i, \text{sim}(q, e_i), m_i) : \text{sim}(q, e_i) \geq \epsilon\}$$

where sim is a similarity function (cosine similarity, Euclidean distance, etc.).

4.5.3 Result Properties

The querier receives, for each match:

- The derived pseudonym pk_i (unique per fragment, unlinkable).
- The similarity score.
- Public metadata m_i.

The querier *cannot* determine from the results:

- Which matching fragments originated from the same document.
- Which matching fragments originated from the same user.
- How many distinct users contributed the matching fragments.
- The content of the underlying raw data.

4.6 Anonymous Relay Communication

When a querier wishes to contact the owner of a matching fragment, the system provides an anonymous relay mechanism (see Figure 6):

4.6.1 Message Submission

The querier composes a message (e.g., an access request, a collaboration proposal, a removal request) and submits it to the relay service addressed to the target fragment's derived pseudonym pk_i. The querier may generate their own one-time derived pseudonym pk_q as a return address, ensuring the querier's identity is also protected.

4.6.2 Relay Routing

The mapping from each derived pseudonym pk_i to the owning user's identity is end-to-end encrypted and stored only on the user's device. The relay service routes messages using encrypted forwarding information that it cannot read—it can deliver a message addressed to pk_i to the correct destination without learning who owns that pseudonym. A compromised relay cannot deanonymize users.

4.6.3 Owner Response

The owner receives the message, sees only the querier's one-time pseudonym pk_q as the sender, and may respond through the same relay mechanism. Neither party learns the other's real identity unless they choose to disclose it.

4.6.4 Enhanced Routing Privacy

To mitigate the risk of the relay service itself linking pseudonyms to owners, the system may employ one or more of the following techniques:

(a) **Per-fragment relay addresses.** Each fragment is assigned a distinct relay intermediary, so that no single relay node sees the forwarding mappings for all of a user's fragments.

(b) **Threshold decryption routing.** The forwarding address for each fragment is encrypted under a threshold scheme requiring k of n relay nodes to collaborate for decryption, preventing any single node from reading the mapping.

(c) **Onion-routed delivery.** Messages are encapsulated in multiple layers of encryption, each layer addressed to a different relay node, such that no single node in the routing path knows both the sender and the final destination.

4.7 Fragment Revocation and Update

A data owner may revoke a previously published fragment from the discovery index by signing a revocation message with the derived private key sk_i corresponding to that fragment's pseudonym pk_i. The discovery index verifies the signature against pk_i and removes the entry from the searchable index. Because only the holder of sk_i can produce a valid signature, revocation is authenticated without revealing the owner's real identity or linking the revoked entry to any other entry.

To update a fragment (e.g., after editing the underlying data), the owner revokes the old entry and publishes a new fragment with a new content hash and therefore a new, unrelated derived pseudonym. The old and new entries are unlinkable.

4.8 Anonymity Properties of the Combined System

The combination of content-aware fragmentation, per-fragment pseudonyms, and independent vectorization produces the following anonymity properties:

4.8.1 Fragment Unlinkability

An observer of the public discovery layer who sees N published fragments sees N independent pseudonyms. The observer cannot determine:

- Whether any two fragments originated from the same document.

- Whether any two fragments originated from the same user.
- The minimum or maximum number of distinct users represented.

For example, if a user publishes an essay fragmented into 50 sentences and a photo fragmented into 12 regions, the 62 resulting entries appear as 62 unrelated entries from 62 potentially distinct users.

4.8.2 Anonymity Set Growth

As the number of users N and entries E in the network grows, the anonymity set for any given fragment grows linearly with N (see Figure 7). An adversary's best guess for the owner of any single fragment is uniform over all N users, because pseudonyms carry no information about their origin. This is the opposite of conventional networks, where more data enables more profiling: in this system, more participants means stronger anonymity.

4.8.3 Elimination of Content-Based Correlation

A conventional pseudonymous system publishes whole documents or complete data records under a pseudonym, providing an adversary with rich content from which to infer authorship through stylistic analysis, topic modeling, or metadata patterns. The fragmentation step eliminates this attack surface.

Because data is shredded to the finest meaningful granularity, individual fragments carry insufficient signal for content-based correlation. A single sentence extracted from an essay contains no reliable authorship fingerprint: it is too short for stylistic analysis, carries no document-level structural information, and is semantically indistinguishable from sentences written by other authors on the same topic. A single overlapping image region extracted from a photograph contains no camera-level signature, no composition information, and no scene-level context. A single data field from a structured record carries no record-level identity.

The shredding operates at a granularity below the threshold at which content-based linkage is meaningful. Combined with per-fragment pseudonyms that are cryptographically unlinkable, the result is that an adversary observing the public discovery layer sees a collection of atomic, unrelated fragments, each under an independent pseudonym, with no basis for grouping them by document or by user.

5 Limitations

- **Embedding inversion.** Vector embeddings preserve semantic similarity, which means they reveal that a fragment is *about* a particular topic. However, the threat from embedding inversion is severely constrained in this system: each fragment is a single sentence or image region under a unique pseudonym, so even a successful inversion yields an isolated, unattributable sentence fragment of uncertain accuracy with no way to verify the reconstruction or link it to other fragments or users.
- **Relay as metadata bottleneck.** While the relay cannot read the pseudonym-to-identity mapping (it is end-to-end encrypted on the user's device), it can observe traffic patterns—which pseudonyms receive messages and when. High-traffic pseudonyms could be correlated with timing of user activity. The enhanced routing options (per-fragment relay addresses, threshold decryption, onion routing) mitigate this.

- **Fragmentation quality.** The privacy guarantees depend on fragments being small enough that individual fragments carry insufficient signal for content-based correlation. Poorly chosen fragmentation boundaries (e.g., entire paragraphs that are stylistically distinctive) could weaken this property.

6 Related Work

Cryptocurrency stealth addresses (e.g., Monero) use per-transaction derived keys to achieve transaction unlinkability, but operate on financial transactions with exact-match lookups rather than semantic similarity search, and do not address fragmentation or anonymous post-discovery communication.

Private information retrieval (PIR) protocols allow querying a database without revealing which record was accessed, but do not address the inverse problem of allowing others to discover that relevant data exists.

Searchable encryption and homomorphic encryption allow queries over encrypted data but typically require a shared encryption scheme and do not support semantic similarity search across unrelated parties.

Federated learning enables distributed model training but does not provide point-query discovery of specific similar entries across a network.

7 Conclusion

We have presented a system that resolves the fundamental tension between data privacy and data discoverability. By fragmenting private data into semantically meaningful segments, independently embedding each fragment, and publishing embeddings under cryptographically unlinkable per-fragment pseudonyms, the system enables similarity-based discovery without exposing content, identity, or fragment associations. The anonymity set grows linearly with network size—more participants means stronger privacy, not weaker. End-to-end encrypted relay routing enables post-discovery communication without either party revealing their identity. The result is a discovery layer where private data can be found without being exposed.

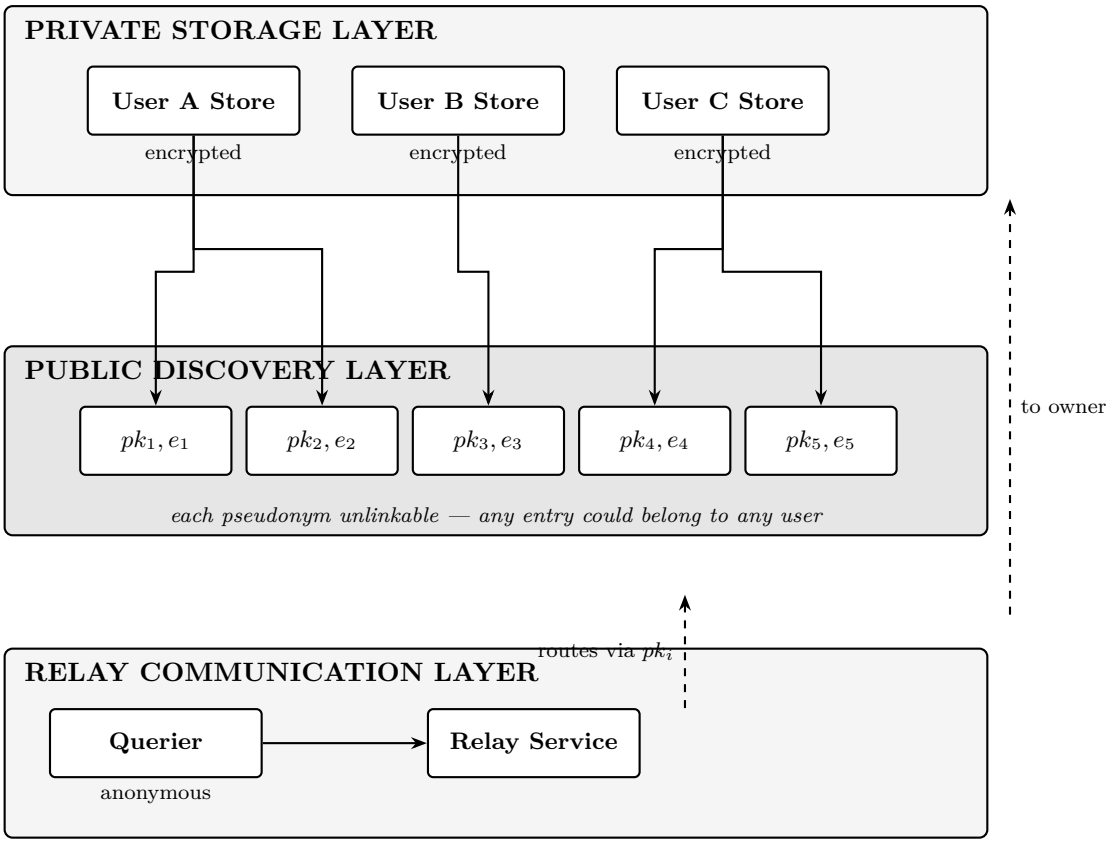


Figure 1: **System Architecture.** Three-layer architecture: private storage (encrypted, user-controlled), public discovery (vectorized fragments with unlinkable pseudonyms), and relay communication (anonymous message routing).

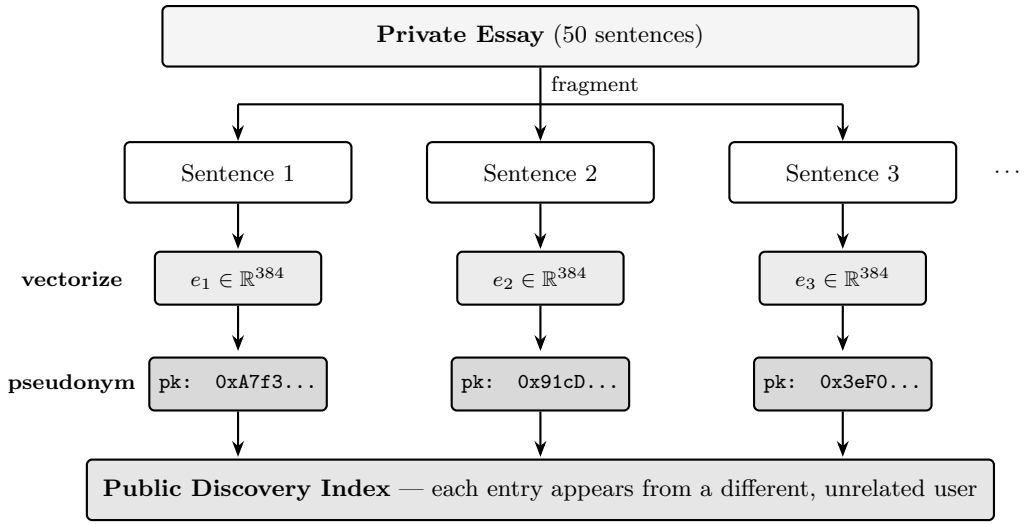


Figure 2: **Text Fragmentation.** An essay is split into individual sentences. Each sentence is independently vectorized and tagged with a unique, unlinkable pseudonym before publication. No two entries can be associated.

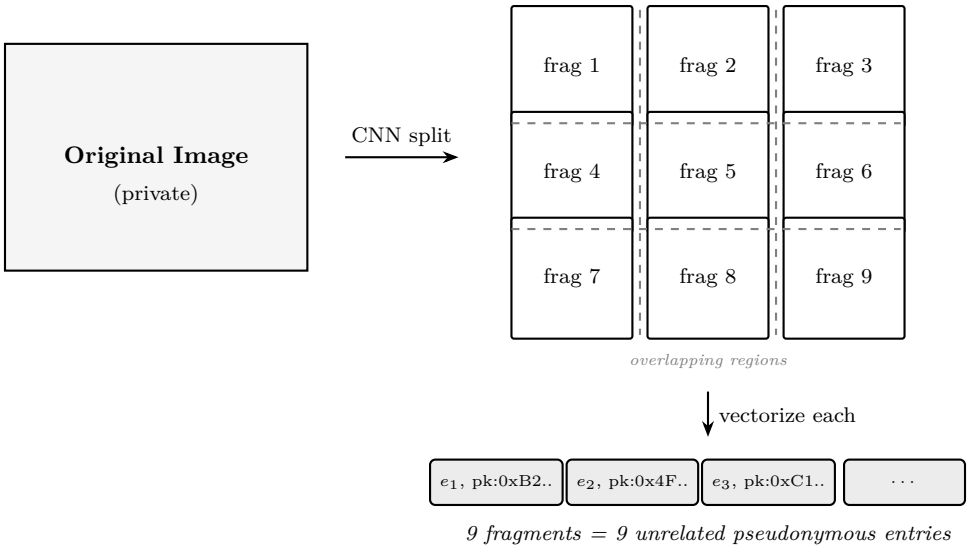
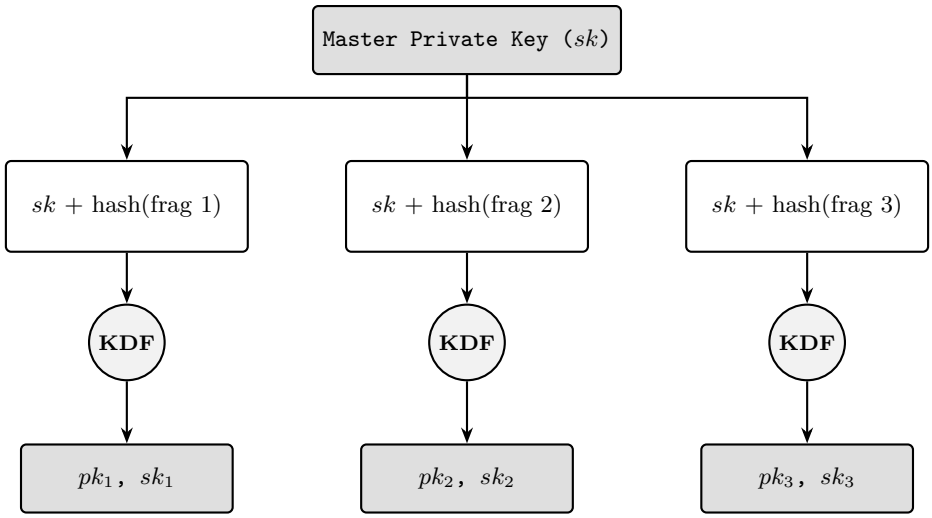


Figure 3: **Image Fragmentation.** A photograph is processed through CNN-based overlapping spatial segmentation. Each region is independently vectorized and published under a unique pseudonym. Overlap ensures features at boundaries are captured.



Each (pk_i, sk_i) is unique and unlinkable. An observer cannot determine that pk_1, pk_2, pk_3 share the same master key. Owner can re-derive any pk_i from sk and h_i.

Figure 4: **Per-Fragment Pseudonym Derivation.** A deterministic key derivation function (KDF) combines the user's master private key with each fragment's content hash to produce unique, unlinkable public/private key pairs. The owner retains the ability to prove ownership.

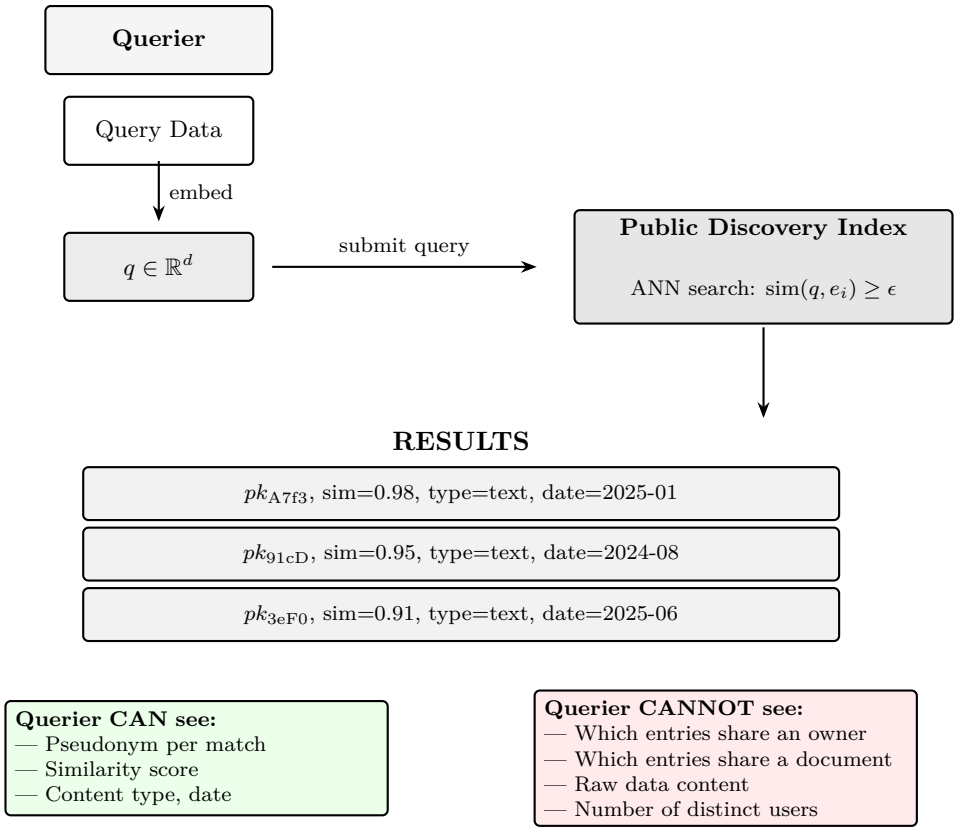
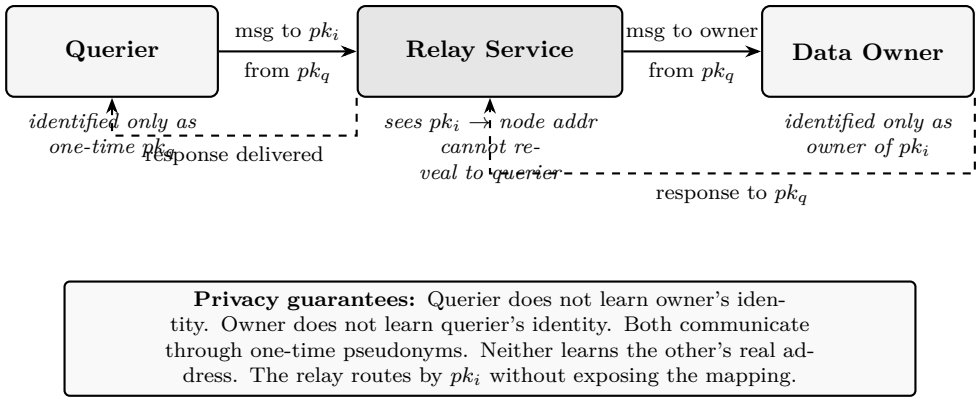


Figure 5: **Similarity Discovery Protocol.** A querier embeds their data, submits the vector to the public index, and receives matching entries identified only by unlinkable pseudonyms and similarity scores. The querier cannot determine associations between entries or access raw data.



Privacy guarantees: Querier does not learn owner's identity. Owner does not learn querier's identity. Both communicate through one-time pseudonyms. Neither learns the other's real address. The relay routes by pk_i without exposing the mapping.

Figure 6: **Anonymous Relay Communication.** A querier sends a message addressed to a fragment's pseudonym. The relay service forwards it to the data owner without revealing the identity mapping. The owner responds through the same relay. Neither party learns the other's real identity.

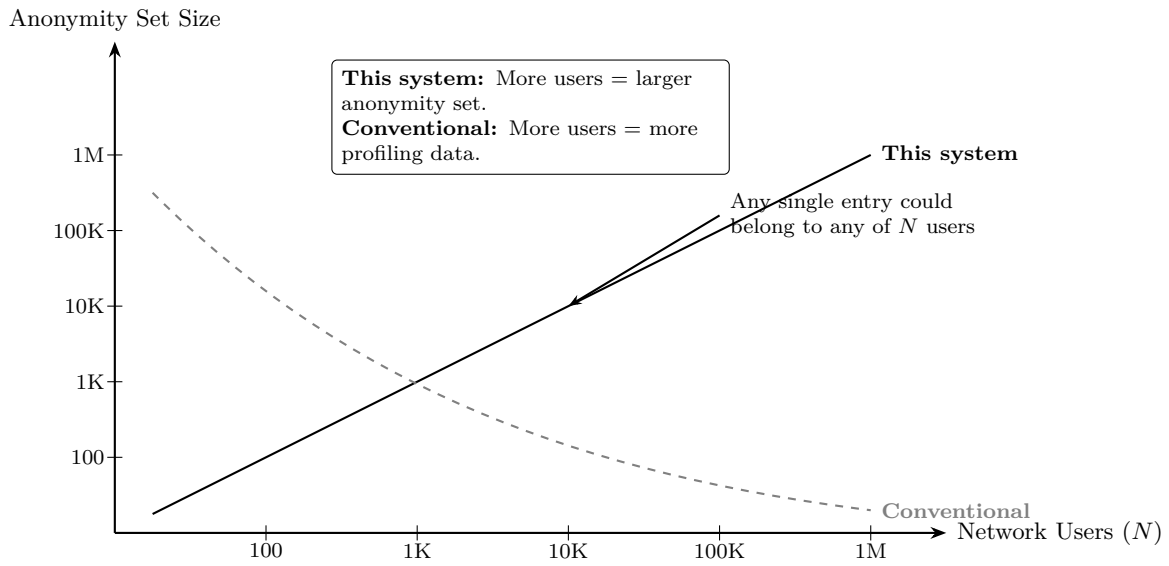


Figure 7: **Anonymity Set Growth.** In this system, the anonymity set for any given fragment grows linearly with the number of network participants. This is the opposite of conventional networks, where growth enables more profiling and deanonymization.